

5 Pineapples contain the sugars fructose, glucose and sucrose.

- (a) (i) Which row of the table shows which of these sugars are monosaccharides and which are disaccharides?

(1)

	Monosaccharides	Disaccharides
☐ A	fructose only	glucose and sucrose
☐ B	glucose only	fructose and sucrose
☐ C	fructose and glucose	sucrose only
☐ D	glucose and sucrose	fructose only

- (ii) Which type of bond joins two monosaccharides together to form a disaccharide?

(1)

- ☐ A ester
- ☐ B glycosidic
- ☐ C peptide
- ☐ D phosphodiester



P 6 1 4 6 9 A 0 1 5 2 8

(b) The table shows the concentration of these sugars in three pineapples.

Sugar	Concentration of sugar / g cm^{-3}		
	Pineapple 1	Pineapple 2	Pineapple 3
fructose	1.71	1.44	1.41
glucose	1.22	1.02	1.00
sucrose	9.08	7.77	8.81

(i) Calculate the mean concentration of glucose in these three pineapples.

Give your answer in g dm^{-3} .

(2)

Answer g dm^{-3}

*(ii) Pineapples can be preserved by a process called osmotic dehydration in which pieces of pineapple are submerged in a solution of sugars.

The diagram shows how this process can be carried out.



